

MIROFISH-OFFLINE

# Complete Setup Guide

Zero Cloud · No API Keys · 100% Local · English UI

Platform

Windows 10 / Windows 11

Experience

Beginner friendly — every step explained

API keys needed

NONE — zero cloud dependencies

Internet after setup

NOT required — runs fully offline

Cost

Free — all software is open source

Time required

45–90 minutes (model download is the longest step)

What runs locally

Ollama (AI brain) + Neo4j (memory) + MiroFish UI

## IMPORTANT

This guide installs everything on your Windows laptop. After setup is complete, MiroFish runs with no internet connection, no API keys, no monthly costs, and no data leaving your machine.

## What We Are Installing

MiroFish-Offline replaces all cloud services with local equivalents. Here is what each piece does:

| Tool             | What It Does                                            | Size    | Cost      |
|------------------|---------------------------------------------------------|---------|-----------|
| Docker Desktop   | Runs Neo4j (the memory database) in a safe container    | ~600 MB | Free      |
| Git              | Downloads the MiroFish-Offline code from GitHub         | ~50 MB  | Free      |
| Node.js          | Runs the MiroFish website interface (frontend)          | ~30 MB  | Free      |
| Python 3.11      | Runs the MiroFish AI pipeline (backend)                 | ~80 MB  | Free      |
| uv               | Python package manager — installs backend libraries     | ~10 MB  | Free      |
| Ollama           | Runs the AI language model 100% locally on your CPU/GPU | ~50 MB  | Free      |
| qwen2.5 model    | The actual AI brain — runs inside Ollama                | 5–20 GB | Free      |
| nomic-embed-text | Converts text to numbers for memory search              | ~274 MB | Free      |
| MiroFish-Offline | The simulation engine itself (English UI fork)          | ~50 MB  | Free/AGPL |

### IMPORTANT

The model download (qwen2.5) is the biggest step — 5 to 20 GB depending on which model you choose. Make sure you are on a good internet connection for that step. Everything else is small.

## Step 0 — Choose Your Model Based on Your RAM

Before starting, look at this table and remember which model you will use. You need to know this for Step 6.

To check your RAM on Windows: press Windows key, type 'About your PC', press Enter. Look for 'Installed RAM'.

| Your RAM  | Model to Use | Download Size | Speed                                 |
|-----------|--------------|---------------|---------------------------------------|
| 8 GB RAM  | qwen2.5:7b   | ~4.7 GB       | Slow but works — recommended for 8GB  |
| 16 GB RAM | qwen2.5:14b  | ~9 GB         | Good quality. Recommended for 16GB    |
| 32 GB RAM | qwen2.5:32b  | ~20 GB        | Best quality. Use if you have the RAM |

### TIP

If you are unsure, start with qwen2.5:7b. It works on any laptop with 8GB RAM. You can always download a bigger model later without reinstalling anything.

### WARNING

DO NOT try to run qwen2.5:32b on a laptop with 8GB RAM — your computer will run out of memory and freeze. Use qwen2.5:7b for 8GB.

## Phase 1 — Install the Four Foundation Tools

These four tools are the foundation. Install them in order. Each one takes 5-10 minutes.

### Step 1 — Open PowerShell as Administrator

You will type all commands into PowerShell. Here is how to open it:

1

Press the Windows key on your keyboard

2

Type: PowerShell

3

Right-click 'Windows PowerShell' in the results and choose 'Run as administrator'

Click Yes if Windows asks permission.

#### WHAT YOU SHOULD SEE

Windows PowerShell  
Copyright (C) Microsoft Corporation.

```
PS C:\Windows\system32> _
```

#### TIP

Keep this PowerShell window open for the entire guide. If you close it by accident, reopen it (Windows key > PowerShell > right-click > Run as administrator) and navigate back with: `cd C:\Users\%USERNAME%\Documents\MiroFish-Offline`

### Step 2 — Install Git

1

Open your browser and go to: <https://git-scm.com/download/win>

The download starts automatically.

2

**Open the downloaded .exe file and click Next on every screen**

Accept all defaults. Click Install. Click Finish.

3

**Close PowerShell completely and reopen it as Administrator**

4

**Type this to confirm Git installed:**

```
git --version
```

#### WHAT YOU SHOULD SEE

```
git version 2.44.0.windows.1
```

#### WARNING

If you see 'git is not recognized' — close PowerShell, reopen as Administrator, and try again. Still failing? Restart your computer.

## Step 3 — Install Node.js

1

**Go to: <https://nodejs.org> — click the big green LTS button**

2

**Open the downloaded .msi file. Click Next on every screen. Click Install.**

3

**Close and reopen PowerShell as Administrator. Then check:**

```
node -v
```

#### WHAT YOU SHOULD SEE

v20.11.0 (any v18, v19, v20, v21, or v22 is fine)

## Step 4 — Install Python 3.11

#### IMPORTANT

MiroFish-Offline needs Python 3.11 specifically. Do NOT install Python 3.13 or 3.12 — they will not work.

1

**Go to: <https://www.python.org/downloads/release/python-3119/>**

Scroll down to 'Files'. Click 'Windows installer (64-bit)'.

2

**Open the downloaded file. BEFORE clicking anything — tick the checkbox at the BOTTOM that says 'Add Python 3.11 to PATH'**

This checkbox is critical. If you miss it, Python will not work.

3

**Click Install Now. Wait for it to finish. Click Close.**

4

**Close and reopen PowerShell as Administrator. Check:**

```
python --version
```

#### WHAT YOU SHOULD SEE

Python 3.11.9

### WARNING

If you see Python 3.13 or 3.12 — uninstall it from Windows Settings > Apps, then redo Step 4 making sure to tick the 'Add Python 3.11 to PATH' checkbox.

## Step 5 — Install uv

Paste this entire command into PowerShell and press Enter:

```
powershell -ExecutionPolicy Bypass -c "irm  
https://astral.sh/uv/install.ps1 | iex"
```

### WHAT YOU SHOULD SEE

Downloading uv...  
Installing to C:\Users\YourName\.local\bin\uv  
uv was installed successfully!

1

### Close PowerShell completely and reopen as Administrator

Then check:

```
uv --version
```

### WHAT YOU SHOULD SEE

uv 0.4.x

### SUCCESS

Phase 1 complete — Git, Node.js, Python 3.11, and uv are all installed.

## Phase 2 — Install Docker Desktop (for Neo4j)

Docker runs Neo4j — the local graph database that gives MiroFish its memory. Docker is a safe, widely used tool that creates isolated containers on your machine.

1

Go to: <https://www.docker.com/products/docker-desktop/>

2

Click 'Download for Windows' and open the installer

3

**During installation — leave all defaults ticked. Click OK. Click Install.**  
When asked to restart your computer — click 'Restart'. This is required.

4

**After restart, Docker Desktop will open automatically**  
You may see a tutorial prompt — you can skip it. Look for the whale icon in your Windows system tray (bottom right) — when it stops animating, Docker is ready.

5

**Open PowerShell as Administrator and check Docker is running:**

```
docker --version
```

### WHAT YOU SHOULD SEE

Docker version 26.1.0, build a5ee5b1

### WARNING

If Docker Desktop shows an error about WSL 2 — click the link it shows to install WSL 2, restart your computer, then reopen Docker Desktop. This is a common first-time step on Windows.

**TIP**

Docker Desktop must be running (whale icon in system tray) every time you use MiroFish-Offline. If you restart your computer, open Docker Desktop before starting MiroFish.

## Phase 3 — Install Ollama (Local AI Engine)

Ollama runs the AI language model entirely on your laptop. No internet, no API key, no cost.

1

Open your browser and go to: <https://ollama.ai>

2

Click 'Download for Windows'

3

Open the downloaded `OllamaSetup.exe` and click Install

4

After installation, Ollama runs in the background automatically

Look for the Ollama icon in your system tray (bottom right of Windows).

5

In PowerShell, check Ollama is running:

```
ollama --version
```

### WHAT YOU SHOULD SEE

```
ollama version 0.3.x
```

### TIP

Ollama must be running (system tray icon visible) every time you use MiroFish-Offline.

## Download the AI Models

Now download the two models Ollama needs. Choose the main LLM model based on your RAM (from Step 0). The embedding model is the same for everyone.

### IMPORTANT

Look at the Step 0 table to know which model you chose. Type the command for YOUR model below. Do NOT run all three — just run the one that matches your RAM.

**For 8 GB RAM — type this (downloads ~4.7 GB):**

```
ollama pull qwen2.5:7b
```

**For 16 GB RAM — type this (downloads ~9 GB):**

```
ollama pull qwen2.5:14b
```

**For 32 GB RAM — type this (downloads ~20 GB):**

```
ollama pull qwen2.5:32b
```

### TIP

This download can take 15-60 minutes depending on your internet speed. The progress bar will show percentages. Do not close PowerShell.

**Everyone downloads this embedding model (downloads ~274 MB):**

```
ollama pull nomic-embed-text
```

### WHAT YOU SHOULD SEE

```
pulling manifest
pulling ..... 100%
success
```

Verify both models downloaded:

```
ollama list
```

WHAT YOU SHOULD SEE

| NAME                    | ID     | SIZE  |
|-------------------------|--------|-------|
| qwen2.5:7b (or 14b/32b) | abc123 | 4.7GB |
| nomic-embed-text        | def456 | 274MB |

SUCCESS

Phase 3 complete — Ollama is installed and your AI models are downloaded.

## Phase 4 — Download MiroFish-Offline

Now we download the actual MiroFish-Offline code from GitHub.

1

Go to your Documents folder in PowerShell:

```
cd C:\Users\%USERNAME%\Documents
```

2

Download MiroFish-Offline:

```
git clone https://github.com/nikmcfly/MiroFish-Offline.git
```

### WHAT YOU SHOULD SEE

Cloning into 'MiroFish-Offline'...  
Receiving objects: 100% (xxx/xxx), done.

3

Go into the MiroFish-Offline folder:

```
cd MiroFish-Offline
```

### WHAT YOU SHOULD SEE

```
PS C:\Users\YourName\Documents\MiroFish-Offline>
```

### TIP

Every command from now on must be run from inside this folder. If you ever close PowerShell

and reopen it, type: `cd C:\Users\%USERNAME%\Documents\MiroFish-Offline` to get back here.

## Phase 5 — Configure MiroFish-Offline

MiroFish-Offline needs a configuration file that tells it where to find Ollama and Neo4j. No API keys needed.

1

Create the .env configuration file:

```
copy .env.example .env
```

2

Open the .env file in Notepad:

```
notepad .env
```

Notepad will open. Find the lines that start with LLM\_API\_KEY, LLM\_BASE\_URL, LLM\_MODEL\_NAME, NEO4J\_URI, and EMBEDDING\_MODEL.

3

Set the values exactly as shown below for your model — replace the placeholders:

**If you downloaded qwen2.5:7b (8 GB RAM):**

```
LLM_API_KEY=ollama
LLM_BASE_URL=http://localhost:11434/v1
LLM_MODEL_NAME=qwen2.5:7b

NEO4J_URI=bolt://localhost:7687
NEO4J_USER=neo4j
NEO4J_PASSWORD=mirofish

EMBEDDING_MODEL=nomic-embed-text
EMBEDDING_BASE_URL=http://localhost:11434
```

---

**If you downloaded qwen2.5:14b (16 GB RAM):**

```
LLM_API_KEY=ollama
LLM_BASE_URL=http://localhost:11434/v1
LLM_MODEL_NAME=qwen2.5:14b

NEO4J_URI=bolt://localhost:7687
NEO4J_USER=neo4j
NEO4J_PASSWORD=mirofish

EMBEDDING_MODEL=nomic-embed-text
EMBEDDING_BASE_URL=http://localhost:11434
```

**If you downloaded qwen2.5:32b (32 GB RAM):**

```
LLM_API_KEY=ollama
LLM_BASE_URL=http://localhost:11434/v1
LLM_MODEL_NAME=qwen2.5:32b

NEO4J_URI=bolt://localhost:7687
NEO4J_USER=neo4j
NEO4J_PASSWORD=mirofish

EMBEDDING_MODEL=nomic-embed-text
EMBEDDING_BASE_URL=http://localhost:11434
```

## 4

**Save the file in Notepad: press Ctrl+S, then close Notepad**

### IMPORTANT

Notice: LLM\_API\_KEY is set to the word 'ollama' — not a real API key. This is correct. It tells MiroFish to use Ollama instead of a cloud service. Do not change it.

## Phase 6 — Start Neo4j (Memory Database)

Neo4j is the graph database that stores agent memories. We run it using Docker.

1

### Make sure Docker Desktop is open and running

Check the whale icon in the Windows system tray. If it is not there, open Docker Desktop from the Start menu and wait for it to finish loading (the whale icon will stop animating).

2

### In PowerShell, start Neo4j with this command:

```
docker run -d --name neo4j-mirofish ^  
-p 7474:7474 -p 7687:7687 ^  
-e NEO4J_AUTH=neo4j/mirofish ^  
neo4j:5.15-community
```

### TIP

Copy this command exactly. The ^ symbols at the end of lines let you write one long command across multiple lines in PowerShell. Paste all lines together.

### WHAT YOU SHOULD SEE

```
Unable to find image 'neo4j:5.15-community' locally  
5.15-community: Pulling from library/neo4j  
...  
Status: Downloaded newer image for neo4j:5.15-community  
a1b2c3d4e5f6g7h8i9j0...
```

3

### Wait 30 seconds, then check Neo4j is running:

```
docker ps
```

### WHAT YOU SHOULD SEE

| CONTAINER ID | IMAGE                | STATUS        | PORTS                  |
|--------------|----------------------|---------------|------------------------|
| a1b2c3d4e5f6 | neo4j:5.15-community | Up 30 seconds | 0.0.0.0:7474->7474/tcp |

4

#### Open your browser and go to: <http://localhost:7474>

This is the Neo4j dashboard. You should see a login screen.

5

#### Login with: Username = neo4j | Password = mirofish

You should see the Neo4j browser interface. You can close this browser tab — you do not need it again.

### SUCCESS

Neo4j is running and storing data. It will continue running in the background until you restart your computer.

### WARNING

If Docker shows an error: make sure Docker Desktop is fully started (whale icon not animating). If you see a port conflict error, stop any previous Neo4j containers: `docker stop neo4j-mirofish` then `docker rm neo4j-mirofish` and run Step 2 again.

## Phase 7 — Install MiroFish Packages

This step downloads all the Python and Node.js libraries MiroFish needs. It runs automatically.

### IMPORTANT

Make sure you are still in the MiroFish-Offline folder. Your prompt should show 'MiroFish-Offline' before the > symbol.

### 1

Install all packages with one command:

```
npm run setup:all
```

This will take 3-15 minutes. You will see a lot of scrolling text. This is normal. Wait until it stops and you see the prompt again.

### WHAT YOU SHOULD SEE

```
Installing Node dependencies...
added 342 packages in 23s
Installing frontend dependencies...
added 891 packages in 45s
Installing Python dependencies...
Installed 67 packages in 38s
PS C:\Users\YourName\Documents\MiroFish-Offline>
```

### WARNING

If you see a Python error: run 'python --version'. It must show 3.11.x. If it shows 3.12 or 3.13, go back and reinstall Python 3.11 (with the PATH checkbox ticked).

### WARNING

If npm is not recognised: close PowerShell, reopen as Administrator, cd back to the folder, and try again.

**SUCCESS**

All packages installed. You are almost ready.

## Phase 8 — Start MiroFish-Offline!

This is the final step. One command starts both the frontend and backend.

1

In PowerShell (inside the MiroFish-Offline folder), type:

```
npm run dev
```

### WHAT YOU SHOULD SEE

```
[backend] * Running on http://127.0.0.1:5001
[backend] Serving Flask app 'run'
[frontend] > vite
[frontend] ready - started server on 0.0.0.0:3000
```

MiroFish-Offline is running!

2

Open your browser (Chrome, Edge, or Firefox)

3

Go to: <http://localhost:3000>

### SUCCESS

You should see the MiroFish-Offline interface in English. It is running 100% on your laptop — no internet, no API keys, no cloud.

### IMPORTANT

Keep the PowerShell window open while using MiroFish. If you close it, MiroFish stops. To stop intentionally: click in PowerShell and press Ctrl+C.

## Phase 9 — Run Your First Simulation

You are inside MiroFish-Offline. Here is how to run your first experiment.

### What is a Seed Document?

A seed document is the information you feed MiroFish. It can be a product launch brief, a news article, a campaign concept, or anything you want simulated consumers to react to.

### Example Seed Document — Copy and Paste This

```
New product launch: 'Aam Zest' zero-sugar mango drink.  
Target: Urban Indians aged 18-35.  
Price: Rs. 50 for 250ml.  
Claims: Real mango pulp, no artificial sweeteners, 40 calories.  
Available: Blinkit, Zepto, and modern retail from June 2026.  
  
Predict: How will Indian consumers respond to this product?  
What are the top objections? Which age group responds best?
```

### Simulation Settings for First Run

| Setting              | Recommended Value | Why                                                |
|----------------------|-------------------|----------------------------------------------------|
| Number of agents     | 20-30             | Small number = faster results on local hardware    |
| Simulation rounds    | 10-15             | Enough for a clear pattern without taking too long |
| Seed document length | 100-250 words     | Enough detail without slowing the model down       |
| Agent diversity      | Mixed (default)   | Let MiroFish decide the agent mix for now          |

#### TIP

On a laptop with 8GB RAM and qwen2.5:7b, a simulation with 20 agents and 10 rounds takes about 15-30 minutes. This is normal — the AI model is doing real work on your CPU. On 16GB+ with a larger model, it is faster.

#### WARNING

Do not close the PowerShell window or your browser during a simulation. The AI is actively running on your CPU. Closing it will cancel the simulation.

## Daily Usage — Starting MiroFish Every Day

After the one-time setup you just completed, starting MiroFish every day takes about 2 minutes and 3 steps.

### Every Time You Want to Use MiroFish

| Step | Action                               | Command / What to Do                                                                |
|------|--------------------------------------|-------------------------------------------------------------------------------------|
| 1    | Start Docker Desktop                 | Open from Start menu. Wait for the whale icon in the system tray to stop animating. |
| 2    | Start Neo4j (if not already running) | In PowerShell: <code>docker start neo4j-mirofish</code>                             |
| 3    | Go to MiroFish folder                | <code>cd C:\Users\%USERNAME%\Documents\MiroFish-Offline</code>                      |
| 4    | Start MiroFish                       | <code>npm run dev</code>                                                            |
| 5    | Open the app                         | Open browser and go to <code>http://localhost:3000</code>                           |

#### IMPORTANT

Ollama starts automatically when Windows starts. You do not need to start it manually. Docker Desktop may or may not start automatically — check the system tray.

### Checking If Services Are Already Running

If you are not sure whether Neo4j is still running from a previous session, type:

```
docker ps
```

If you see `neo4j-mirofish` in the list — it is already running. Skip Step 2. If the list is empty — run:

```
docker start neo4j-mirofish
```

### Stopping MiroFish at End of Day

- Click inside the PowerShell window and press `Ctrl+C` to stop MiroFish

- Docker and Ollama will keep running in the background — this is fine
- Next time you want to use MiroFish, just run: `npm run dev`

## Troubleshooting

If something is not working, find your error below:

| Problem                                         | What To Do                                                                                                                              |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Browser shows blank page at localhost:3000      | Check PowerShell for red error text. The backend may have failed to start. Run: <code>npm run dev</code> again.                         |
| 'npm is not recognized'                         | Close PowerShell. Reopen as Administrator. <code>cd</code> back to MiroFish-Offline folder. Try again.                                  |
| 'python is not recognized'                      | Reinstall Python 3.11. On the first installer screen, TICK 'Add Python 3.11 to PATH'.                                                   |
| Docker error: 'Cannot connect to Docker daemon' | Docker Desktop is not running. Open it from the Start menu. Wait for the whale icon to stop animating.                                  |
| Neo4j container not found                       | Run: <code>docker start neo4j-mirofish</code> . If it says container not found, redo Phase 6 Step 2.                                    |
| Ollama error: 'model not found'                 | Run: <code>ollama list</code> to see downloaded models. If your model is missing, run: <code>ollama pull qwen2.5:7b</code> (or 14b/32b) |
| Simulation is very slow                         | Normal for CPU-only inference. On 8GB RAM with 7b model, 20 agents and 10 rounds = 15-30 mins. This is expected.                        |
| Computer freezes during simulation              | Your RAM is running out. Reduce agents to 10 and rounds to 5. Or switch to <code>qwen2.5:7b</code> in your <code>.env</code> file.      |
| 'uv' is not recognized                          | Close PowerShell, reopen as Administrator. <code>uv</code> needs a fresh session to be recognised.                                      |
| Backend starts but frontend fails               | Run: <code>cd frontend &amp;&amp; npm install &amp;&amp; cd ..</code> and then <code>npm run dev</code> again.                          |
| Port 3000 already in use                        | Restart your computer. Then open Docker Desktop, <code>cd</code> to MiroFish-Offline folder, and run <code>npm run dev</code> .         |
| Neo4j login fails at localhost:7474             | Make sure password is 'mirofish' (lowercase). If you changed it, update <code>NEO4J_PASSWORD</code> in your <code>.env</code> file.     |

### Still stuck?

GitHub Issues page: <https://github.com/nikmcfly/MiroFish-Offline/issues>

Create a new issue and paste the exact error message from your PowerShell window.

The community is active and typically responds within hours.

## Quick Reference Card

Print or screenshot this page for easy daily reference.

### Daily Commands

| What                      | Command                                                        |
|---------------------------|----------------------------------------------------------------|
| Go to MiroFish folder     | <code>cd C:\Users\%USERNAME%\Documents\MiroFish-Offline</code> |
| Start Neo4j               | <code>docker start neo4j-mirofish</code>                       |
| Check if Neo4j is running | <code>docker ps</code>                                         |
| Start MiroFish            | <code>npm run dev</code>                                       |
| Stop MiroFish             | Ctrl+C in PowerShell                                           |
| Check Ollama models       | <code>ollama list</code>                                       |
| Pull a new model          | <code>ollama pull qwen2.5:7b</code> (or 14b or 32b)            |
| Update MiroFish-Offline   | <code>git pull</code> then <code>npm run setup:all</code>      |

### Service URLs

| Service                        | Address in Browser                  |
|--------------------------------|-------------------------------------|
| MiroFish app (main UI)         | <code>http://localhost:3000</code>  |
| MiroFish backend API           | <code>http://localhost:5001</code>  |
| Neo4j browser (database admin) | <code>http://localhost:7474</code>  |
| Ollama API (check it is live)  | <code>http://localhost:11434</code> |

### Your Configuration Summary

| Setting        | Value                                     |
|----------------|-------------------------------------------|
| LLM_API_KEY    | ollama (not a real key — this is correct) |
| LLM_BASE_URL   | <code>http://localhost:11434/v1</code>    |
| LLM_MODEL_NAME | qwen2.5:7b OR qwen2.5:14b OR qwen2.5:32b  |
| NEO4J_URI      | <code>bolt://localhost:7687</code>        |
| NEO4J_USER     | neo4j                                     |
| NEO4J_PASSWORD | mirofish                                  |

|                    |                        |
|--------------------|------------------------|
| EMBEDDING_MODEL    | nomic-embed-text       |
| EMBEDDING_BASE_URL | http://localhost:11434 |

**MiroFish-Offline is running entirely on your machine.**

No cloud. No API keys. No data leaving your laptop. Every simulation runs 100% locally.

*MiroFish-Offline Windows Setup Guide | 2026*